

Technical Summary:

Mirror-Consistency: Harnessing Inconsistency in Majority Voting

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1 Research Problem and Motivation

1.1 Problem

Sample-based decoding is widely used to improve Large Language Model (LLM) reasoning by generating multiple candidate answers and selecting a final answer via plurality or majority voting. A prominent baseline is Self-Consistency, which repeatedly samples reasoning traces and returns the most frequent answer. The central problem addressed is that plurality voting discards minority answers, even when those disagreements expose uncertainty and potential error patterns.

1.2 Motivation and Gap

Minority answers are treated as noise under standard voting, which creates two practical gaps. First, reasoning accuracy can be limited because informative disagreements are not exploited to correct subsequent generations. Second, confidence estimation based on agreement patterns can become overconfident because the aggregation mechanism ignores the reasons for disagreement. Mirror-Consistency is motivated by the premise that disagreement across sampled answers is an actionable signal rather than a byproduct of sampling.

1.3 Goal

The goal is to improve both (i) reasoning accuracy under comparable sampling budgets and (ii) sample-based confidence calibration by explicitly reflecting on inconsistencies during the decoding process and using the reflection to steer later samples.

2 Related Work

2.1 Self-Consistency and Voting-Based Decoding

Self-Consistency improves reasoning by sampling multiple chains of thought and marginalizing to the most frequent answer, but the aggregation rule focuses on the single most frequent outcome and does not operationalize minority viewpoints.

Table 1: Datasets, task types, and split settings used in the experiments.

Dataset	Task Type	Split Setting Used in Experiments
GSM8K	Arithmetic reasoning	First 768 examples from the test split
SVAMP	Arithmetic reasoning	First 768 examples from the test split
StrategyQA	Multi-hop QA	First 490 test samples
Date Understanding	Multi-hop QA	Full split

2.2 Iterative Refinement and Reflection

Iterative prompting and self-reflection methods attempt to refine an initial answer through feedback. Mirror-Consistency differs by grounding feedback in contrastive evidence, namely discrepancies between a newly generated answer and the current majority vote, rather than reflecting on a single answer in isolation.

2.3 Sample-Based Confidence Calibration

Sample-based calibration methods estimate confidence using properties of the answer distribution, such as agreement with the majority answer. Mirror-Consistency contributes a reflective sampling process intended to better quantify uncertainty, particularly in overconfident regimes, by repeatedly comparing and reflecting on disagreements.

3 Dataset Construction

3.1 Dataset Sources and Scope

No new dataset is constructed. Experiments are conducted on four established benchmarks spanning arithmetic reasoning and multi-hop question answering.

3.2 Dataset Characteristics

GSM8K consists of elementary-level math word problems that test arithmetic and logical reasoning. SVAMP stresses robustness to changes in phrasing and problem structure. StrategyQA requires multi-step inference to answer questions with Boolean outputs. Date Understanding evaluates numerical and chronological reasoning by requiring date calculations from relative time descriptions.

4 Query Protocol and Task Definitions

4.1 Input, Output, and Sampling Protocol

For each input query x (a problem statement or question), the decoding protocol generates multiple candidate outputs, each containing a final answer (often produced via Chain-of-Thought prompting). Two counters are used to characterize cost:

- **#Ans**: the number of candidate answers generated for voting.
- **#Call**: the number of LLM invocations required to complete the protocol.

4.2 Voting and Notation

Let $\hat{r}_1, \dots, \hat{r}_n$ denote n candidate answers sampled for the same input x . Let $\mathbb{I}(\cdot)$ denote an indicator function that equals 1 if its argument is true and 0 otherwise. The most-voted answer is defined by majority voting as follows.

$$\bar{r} = \arg \max_r \sum_{i=1}^n \mathbb{I}(\hat{r}_i = r) \quad (1)$$

Equation (1) defines the majority-vote answer $\bar{r}$. Here, r ranges over possible answer strings, n is the number of sampled answers, and $\hat{r}_i$ is the i th sampled answer. The term $\mathbb{I}(\hat{r}_i = r)$ counts whether the i th sample equals r . The summation therefore counts the frequency of r among the samples, and the $\arg \max$ selects the most frequent answer. This equation is used both for final answer selection and as a reference point for detecting inconsistencies.

5 Modeling Approach

5.1 Core Idea

Mirror-Consistency augments Self-Consistency with a reflective loop that detects when a newly sampled answer disagrees with the current majority vote, produces feedback about the discrepancy, and then conditions subsequent sampling on a growing checklist of feedback items.

5.2 Mirror-Consistency Procedure

Mirror-Consistency uses three prompt components:

- p_{sample} : prompt for initial sampling.
- p_{contrast} : prompt for reflecting on inconsistency between the current majority vote and a newly sampled answer.
- $p_{\text{sample_fb}}$: prompt for conditional resampling using accumulated feedback.

Concatenation of prompt segments and inputs is denoted by $\parallel$ and represents string concatenation in the prompt.

Algorithmically, the method maintains:

- r_k : the k th sampled response.
- m_k : the majority vote computed from $\{r_0, \dots, r_k\}$.
- C_k : a checklist of feedback items accumulated up to iteration k .
- K : the number of reflective iterations.

The procedure begins with an initial response r_0 and initializes $C_0 = \emptyset$ and $m_0 = r_0$. For each iteration $k = 1, \dots, K$, a new response r_k is conditionally sampled using C_{k-1} , then a new checklist C_k is produced by contrasting r_k with m_{k-1} , and finally the majority vote m_k is updated from all sampled responses so far. The final output is m_K .

5.3 Confidence Metrics for Calibration

Mirror-Consistency is also used as a generator of answer distributions for sample-based confidence estimation. The following confidence metrics are defined for a given input x using n sampled answers and the majority vote $\bar{r}$.

$$\text{Agree}(\bar{r}) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(\hat{r}_i = \bar{r}) \quad (2)$$

Equation (2) defines agreement-based confidence $\text{Agree}(\bar{r})$. The symbol $\bar{r}$ is the most-voted answer from Equation (1), and n is the number of sampled answers. The summation counts how many samples match $\bar{r}$, then normalizes by n to produce a fraction in $[0, 1]$. This quantity is used as a confidence score, where higher agreement corresponds to higher estimated confidence.

Let $\bar{r}_1$ and $\bar{r}_2$ denote the top-1 and top-2 most frequent answers among the samples. First-Second-Distance (FSD) is defined as:

$$\text{FSD}(\bar{r}) = \text{Agree}(\bar{r}_1) - \text{Agree}(\bar{r}_2) \quad (3)$$

Equation (3) defines $\text{FSD}(\bar{r})$ as the gap between the agreement rates of the most frequent and second most frequent answers. $\bar{r}_1$ is the majority-vote answer and $\bar{r}_2$ is the runner-up. A larger gap indicates a more dominant majority answer, which is treated as a stronger confidence signal.

Additional metrics are analyzed by defining the set of unique answers obtained after removing duplicates from $\{\hat{r}_1, \dots, \hat{r}_n\}$. Let this set be denoted $\hat{\mathcal{R}}$ and let $|\hat{\mathcal{R}}|$ be its cardinality. Let p_i be the normalized frequency of the i th unique answer in $\hat{\mathcal{R}}$, so that $\sum_{i=1}^{|\hat{\mathcal{R}}|} p_i = 1$.

$$\text{Ent}(\bar{r}) = 1 - \left(-\frac{1}{\log |\hat{\mathcal{R}}|} \sum_{i=1}^{|\hat{\mathcal{R}}|} p_i \log p_i \right) \quad (4)$$

Equation (4) defines an entropy-based consistency score $\text{Ent}(\bar{r})$. The inner term $\sum_i p_i \log p_i$ is the Shannon entropy (up to sign), which measures dispersion of the answer distribution. The normalization factor $\log |\hat{\mathcal{R}}|$ scales the entropy to the range $[0, 1]$ under a uniform distribution over $|\hat{\mathcal{R}}|$ unique answers. The outer transformation $1 - (\cdot)$ inverts the normalized entropy so that higher values correspond to more concentrated, and therefore more consistent, answer distributions.

$$\text{Ans-Num}(\bar{r}) = 1 - \frac{|\hat{\mathcal{R}}|}{n} \quad (5)$$

Equation (5) defines an answer-number-based metric $\text{Ans-Num}(\bar{r})$. Here, $|\hat{\mathcal{R}}|$ is the number of unique answers among n samples. The ratio $|\hat{\mathcal{R}}|/n$ increases when samples are diverse. The subtraction from 1 makes the metric larger when fewer unique answers occur, which corresponds to higher consistency.

Let n_j denote the count of the j th unique answer in $\hat{\mathcal{R}}$. Let $i_{\bar{r}}$ denote the index in $\hat{\mathcal{R}}$ corresponding to the majority answer $\bar{r}$, and let $n_{i_{\bar{r}}}$ be its count. Pairwise comparison consistency is defined as:

$$\text{Pairwise}(\bar{r}) = \prod_{j \neq i_{\bar{r}}} \frac{n_{i_{\bar{r}}}}{n_{i_{\bar{r}}} + n_j} \quad (6)$$

Equation (6) compares the majority answer against each competing unique answer. For each competitor index $j \neq i_{\bar{r}}$, the fraction $\frac{n_{i_{\bar{r}}}}{n_{i_{\bar{r}}} + n_j}$ measures how dominant the majority count is relative to that competitor. The product aggregates dominance across all competitors, yielding a score in $(0, 1]$ where larger values indicate stronger majority dominance and higher consistency.

6 Empirical Results

6.1 Experimental Design

Two evaluation axes are reported.

- **Reasoning accuracy:** measured as task accuracy on GSM8K, SVAMP, StrategyQA, and Date Understanding.
- **Confidence calibration:** measured using Expected Calibration Error (ECE) with confidence scores derived from agreement-based metrics such as Agree and FSD.

Baselines include Standard Chain-of-Thought prompting with a single sample, and Self-Consistency with 10 and 20 samples. Fairness is assessed under two scenarios: equal number of responses (10 vs. 10) and comparable number of calls (19 vs. 20). A fixed sampling temperature $T = 0.4$ is used. For open-source models, inference is conducted on NVIDIA A100 80GB GPUs, with the 70B model requiring two GPUs per inference and the 8B model requiring one GPU.

6.2 Expected Calibration Error

ECE is computed by binning predictions into M confidence intervals and comparing average accuracy to average confidence within each bin. Let B_m denote the set of examples whose confidence scores fall into bin m . Let $|B_m|$ be the number of examples in bin m , and let N be the total number of evaluated examples. Let $\text{acc}(B_m)$ be the empirical accuracy averaged over examples in B_m , and let $\text{conf}(B_m)$ be the average confidence score over the same set.

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (7)$$

Equation (7) defines Expected Calibration Error. M is the number of bins, set to 10 in the reported setup. The term $|\text{acc}(B_m) - \text{conf}(B_m)|$ measures the mismatch between correctness frequency and predicted confidence within bin m . The weight $\frac{|B_m|}{N}$ ensures bins with more examples contribute proportionally more. Lower ECE indicates better calibration because confidence aligns more closely with observed correctness.

6.3 Main Accuracy Results

Table 2 reports accuracy for four LLMs under Standard Chain-of-Thought, Self-Consistency, and Mirror-Consistency. Mirror-Consistency uses #Ans = 10 and #Call = 19. Self-Consistency uses either #Ans = 10 and #Call = 10 or #Ans = 20 and #Call = 20.

The reported results show that Mirror-Consistency improves over Self-Consistency in most settings, including cases where Self-Consistency doubles its number of sampled answers. The strongest gains are observed on arithmetic reasoning benchmarks, where reflection on inconsistencies can correct computation and reasoning mistakes.

6.4 Calibration Results

Table 3 reports ECE for two models on GSM8K and SVAMP under Self-Consistency and Mirror-Consistency, using confidence scores computed by Agree and FSD.

Across both confidence metrics, Mirror-Consistency yields lower ECE than Self-Consistency, indicating improved alignment between confidence scores derived from answer consistency and empirical correctness. The reported calibration curves further indicate reduced overconfidence under Mirror-Consistency.

Table 2: Reasoning accuracy comparison. #Ans is the number of sampled answers. #Call is the number of model invocations.

Model	Method	#Ans	#Call	GSM8K	SVAMP	StrategyQA	Date
GPT-3.5-turbo	Standard CoT	1	1	75.9	77.3	73.6	67.3
GPT-3.5-turbo	Self-Consistency	10	10	85.2	84.2	77.9	72.4
GPT-3.5-turbo	Self-Consistency	20	20	85.4	84.5	77.8	73.0
GPT-3.5-turbo	Mirror-Consistency	10	19	86.7	86.1	78.4	72.6
Qwen-turbo	Standard CoT	1	1	74.5	76.9	71.9	62.7
Qwen-turbo	Self-Consistency	10	10	77.4	79.1	73.8	63.5
Qwen-turbo	Self-Consistency	20	20	77.9	79.1	73.6	63.5
Qwen-turbo	Mirror-Consistency	10	19	79.7	83.0	74.7	64.3
Llama3-8B	Standard CoT	1	1	78.7	80.8	71.6	76.9
Llama3-8B	Self-Consistency	10	10	85.9	89.0	72.0	83.8
Llama3-8B	Self-Consistency	20	20	86.5	89.6	72.8	84.1
Llama3-8B	Mirror-Consistency	10	19	87.2	90.4	72.8	83.5
Llama3-70B	Standard CoT	1	1	92.7	91.8	72.9	90.6
Llama3-70B	Self-Consistency	10	10	95.3	93.5	73.7	94.2
Llama3-70B	Self-Consistency	20	20	95.6	94.0	73.7	94.2
Llama3-70B	Mirror-Consistency	10	19	95.8	93.2	74.7	94.2

Table 3: Expected Calibration Error (ECE). Lower is better. G denotes GSM8K and S denotes SVAMP.

Model	Method	Agree (G)	Agree (S)	FSD (G)	FSD (S)
GPT-3.5-turbo	Self-Consistency	0.081	0.044	0.170	0.127
GPT-3.5-turbo	Mirror-Consistency	0.039	0.042	0.138	0.082
Qwen-turbo	Self-Consistency	0.154	0.196	0.127	0.173
Qwen-turbo	Mirror-Consistency	0.102	0.103	0.089	0.100

7 Summary

7.1 Key Contributions Versus Prior Methods

Mirror-Consistency differs from Self-Consistency by incorporating a reflective mechanism that operationalizes minority disagreements. Instead of treating each sample as independent, the method compares new samples with the current majority vote, generates feedback in the form of a checklist, and conditions later sampling on that checklist. This introduces a structured depth component into an otherwise breadth-oriented sampling procedure.

7.2 Limitations and Future Directions

Several limitations are identified:

- Mirror-Consistency requires sequential sampling because each new answer must be compared with the current majority vote, whereas Self-Consistency can sample in parallel.

- The experiments use Chain-of-Thought as the only prompting strategy for sampling, leaving interactions with other prompting strategies untested in the main setup.
- The approach relies on sufficient diversity among sampled answers; limited diversity reduces the availability of informative inconsistencies.
- The temperature is fixed, and robustness under varying temperatures is not established in the reported configuration.
- Strong instruction-following ability is required because reflection and checklist conditioning are prompt-driven; weaker models can reduce effectiveness.
- Iterative resampling and reflection introduce additional computational cost and associated environmental impact considerations.

Future work directions described include exploring broader prompting strategies, generating prompts that induce diversity across dimensions, varying sampling temperature, and improving efficiency to mitigate sequential and cost constraints.